

AMENDMENT NO. 1 AUGUST 2010
TO
IS 12381 : 1994 COAL (CHAR) REACTIVITY
FOR DIRECT REDUCTION PROCESS —
METHOD OF DETERMINATION

(First Revision)

(Page 2, clause 6.1) — Substitute the following for the existing matter:

‘When sample weight is 5 g and carbon dioxide gas is passed at the rate of 50 c.c./min for 120 minutes then:

$$\begin{aligned}\text{‘Reactivity} &= \frac{W \times 44\,800 \times (1\,050 + 273)}{12 \times 273 \times 120 \times 60 \times 5 \times C_{\text{fix}}} \\ &= \frac{2.5\,128 \times W}{5 \times C_{\text{fix}}} \text{ c.c./g-sec’}\end{aligned}$$

where

C_{fix} = fraction of the Fixed Carbon present in Char before reaction,
and

W = weight loss.’

(Page 3, clause 6.4) — Substitute the following for the existing matter:

‘For the purpose of this test the following conditions shall be maintained:

- a) Temperature : $1\,050 \pm 5^{\circ}\text{C}$
- b) Test duration : 120 minutes
- c) Gas composition : Pure and dry CO_2 gas
- d) Gas flow rate : 50 c.c./min’

(Page 3, clause 6.6, line 8) — Substitute ‘ $1\,050 \pm 5^{\circ}\text{C}$ ’ for ‘ $1\,000 \pm 5^{\circ}\text{C}$ ’.

(Page 3, clause 6.6, line 9 and 10) — Substitute ‘50 c.c./min for 120 minutes’ for ‘100 c.c./min for 25 minutes’ and substitute ‘120 minutes’ for ‘25 minutes’.

Amend No. 1 to IS 12381 : 1994

(Page 3, clause 6.7) — Substitute the following for the existing:

‘Reactivity is reported as follows (see 6.1):

$$\text{Reactivity} = \frac{2.5128 \times W}{5 \times C_{\text{fix}}} \text{ c.c./g-sec'}$$

(MTD 30)